Experiment No.: 6

1. Aim: - Write a program for Time-Series Forecasting with the LSTM Model.

2. **Objective: -** To design and implement a Long Short-Term Memory (LSTM) neural network model for forecasting future values in a time-series dataset.

3. Outcome: - After successful completion of this experiment, students will be able to:

- Understand the concept of time-series forecasting.
- Apply LSTM neural networks for sequential data prediction.
- Analyze and visualize predicted and actual values of time-series data

- **Software Required:** Python

4. Theory:

Time-Series Forecasting is the process of predicting future values based on previously observed data points. It is widely used in areas such as stock price prediction, weather forecasting, sales prediction, and demand forecasting.

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential data. Traditional RNNs suffer from the vanishing gradient problem, which makes learning long sequences difficult. LSTM overcomes this problem using special memory cells and gates.

Components of LSTM

- **Input Gate:** Controls the input flow into the memory cell.
- **Forget Gate:** Determines what information should be removed from memory.
- **Output Gate:** Controls the output from the memory cell.

LSTM networks are highly effective for time-series prediction because they can remember patterns over long-time intervals.

5. **Design / Algorithm**

1. Import necessary libraries (NumPy, Pandas, TensorFlow/Keras, Matplotlib).
2. Load the time-series dataset.
3. Preprocess the dataset and normalize the values.
4. Convert data into training and testing sequences.
5. Build the LSTM model using Keras.

6. Train the model with training data.
7. Predict the future values using the trained model.
8. Visualize the predicted and actual values using graphs.

6. Procedure

Step 1: Import Libraries

Step 2: Load and Preprocess the Dataset

Step 3: Split Data into Training and Testing Sets

Step 4: Prepare Data for LSTM (Create Sequences)

Step 5: Build the LSTM Model

Define the architecture using Keras Sequential model.

Step 6:- Train the Model

Step 7: Make Predictions and Evaluate

Step 8: Visualize the Results

7. Conclusion:-

In this experiment, a time-series forecasting model was implemented using the LSTM neural network. The model was trained using historical data and successfully predicted future values. LSTM networks proved effective in capturing temporal dependencies and patterns in sequential data, making them suitable for time-series prediction tasks.

8. References:-

1. **T. M. Mitchell**, *Machine Learning*, New York, NY, USA: McGraw-Hill, 1997.
2. **K. P. Murphy**, *Machine Learning: A Probabilistic Perspective*, Cambridge, MA, USA: MIT Press, 2012.
3. **G. James, D. Witten, T. Hastie, and R. Tibshirani**, *An Introduction to Statistical Learning with Applications in R*, New York, NY, USA: Springer, 2013.